

EKATA SINGH

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SUMMARY

Final-year B.Tech student with hands-on software engineering experience building enterprise-grade, AI-powered systems and a strong foundation in Data Structures, Algorithms, and problem-solving (Codeforces Expert). Proficient in C++, C, Python, and Java, with a track record of shipping high-quality software and leveraging AI tools to streamline workflows and support decision-making.

EDUCATION

Indira Gandhi Delhi Technical University for Women (IGDTUW) 2022 - 2026
B.Tech, Electronics & Communication Engineering CGPA: 8.70
Kendriya Vidyalaya No. 2, Nau Sena Bagh (CBSE) — Class XII: 95.4% (2021) | Class X: 94.6% (2019)

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Java, SQL
Web & Frameworks: React.js, JavaScript, HTML, CSS, Flask
AI / LLM Tooling: LangChain, LangGraph, RAG Pipelines, Vision RAG, FAISS Vector Database
Developer Tools: Git, GitHub, GitLab, Playwright, REST APIs
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Nomura Jan 2026 – Jun 2026
IT Intern — Quality Assurance & Automation

- Engineered an AI-powered multi-agent system to automate SDLC release verification, validating Jira change requests, release tickets, and mandatory documentation against enterprise governance policies and significantly reducing manual QA effort.
- Developed an LLM-powered Test Case Generation Workbench from scratch that produces functional test cases from Jira Keys and Release Fix Versions, implementing Context Generation and Test Case Generation agents via LangGraph workflows.
- Built an enterprise RAG pipeline that ingests Jira and Confluence documentation into a FAISS vector database, applying query expansion, dense-sparse retrieval, document reranking, and Vision RAG to surface the most relevant context for AI agents.
- Delivered a React analytics dashboard tracking workbench adoption across divisions and business units, and automated the IDATEN credit application in Playwright using reusable scripts, step definitions, and locator-based UI tests.

Tech: Python, Flask, React.js, Playwright, LangChain, LangGraph, RAG Pipelines, FAISS, REST APIs, Jira, Confluence, Git.

PROJECTS

RTL DeepPath Predictor Feb 2025 – Mar 2025
Google Girl Hackathon 2025

- Developed an AI tool that cut RTL synthesis iterations by 70% and achieved 86.67% accuracy in predicting combinational logic depth within ± 1 level, addressing lengthy hardware design cycles.
- Improved model reliability by 18% in R^2 score ($0.5119 \rightarrow 0.8375$) by expanding the training dataset with 60+ complex RTL modules across 6 hardware domains for more robust predictions.

Memory Match Mania Sep 2023 – Dec 2023

- Built a browser-based card-matching game in vanilla JavaScript using the Fisher-Yates shuffle to guarantee fully randomized layouts on every restart.
- Optimized rendering with event delegation and dynamic DOM updates, cutting active event listeners by 80%.

Tech: HTML, CSS, JavaScript.

ACHIEVEMENTS

- Codeforces** — Expert, maximum rating 1658.
- 1st Rank, Reverse Coding** — Tesseract'26 (IIITV Coding Club & Mathematics Club).
- Finalist**, Google Girl Hackathon 2025.
- Selected**, Amazon ML Summer School 2025 — mentorship under Amazon scientists.
- Semi-Finalist**, Flipkart GRiD 6.0 — Software Development Track.

LEADERSHIP & ACTIVITIES

- PR Head, TNP Cell (IGDTUW):** Led placement communication, engaging 20+ recruiting companies.
- Mentor, Desh Ke Mentor:** Mentored 10+ high-school students.